

NBA/NHL Expected Points and Points Over Expected Analysis

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Title and Abstract

NBA/NHL Expected Points and Points Over Expected Analysis. We combine two sports shot datasets to study how shot quality and shot outcomes differ across players. NBA data come from our existing optical-tracking pipeline (2014-15 through 2025-26), and NHL data come from teammate-collected MoneyPuck shot-level records (2007-2024) loaded from a local CSV with portable fallback paths. If data collection is complete as expected, we will build comparable expected-value frameworks (xFG for NBA, xG for NHL), then evaluate player performance with residual-style metrics such as Points/Goals Over Expected and a Shot Difficulty Index. The project goal is to produce a reproducible, side-by-side workflow that identifies players who overperform or underperform shot quality expectations, and to communicate those findings with one clear table, one preliminary figure, and an implementation-ready analysis plan.

Introduction

We are interested in separating shot quality from raw scoring outcomes, since volume and role can hide true efficiency. We already have usable data in both leagues and can demonstrate collected fields immediately (Table 1). The cross-league framing is useful because both sports have location/context effects, but each has different shot environments; comparing expected vs. observed conversion lets us evaluate efficiency in a way that is more fair than raw percentages alone.

Table 1: Sample NBA Player Data (evidence of collected data)

Table 1: Table 1. Sample NBA player-level fields used in our expected-points pipeline.

Player	Shots	FG%	Total POE	POE/100
AJ Green	391	0.4398977	73.166894	18.7127606
Aaron Gordon	273	0.5091575	14.497316	5.3103721
Aaron Nesmith	370	0.3891892	-30.114935	-8.1391715
Aaron Wiggins	363	0.4435262	-3.078856	-0.8481698
Ace Bailey	518	0.4498069	-42.170340	-8.1409924
Ajay Mitchell	456	0.4868421	22.561261	4.9476449

Methods

Data collection is already underway: NBA files are pulled from the established team pipeline, and NHL shot data are read from `shots_2007-2024.csv` with environment-variable and relative-path fallbacks for portability. Our primary analysis tools are R (`readr`, `ggplot2`, `knitr`) plus model outputs from league-specific workflows. We will fit/consume expected-probability models (NBA xFG logistic framework and NHL xG/LASSO outputs), compute player-level residual metrics (actual minus expected scoring), and report difficulty-adjusted comparisons using SDI-style features (distance, angle, rebound/context flags). Key anticipated challenges are path portability, large NHL file size, and balancing model comparability between

leagues; to address this, we use fail-fast path checks, cached NHL chunks, and a consistent reporting template across both sports.

Expected Outcomes

If everything goes as planned, we expect to identify players in both leagues whose observed scoring rates differ meaningfully from expected rates after adjusting for shot difficulty. We also expect to show that high shot volume alone does not imply high efficiency, and that residual-style metrics provide clearer player evaluation than raw conversion percentages.

Preliminary Figure (NHL)

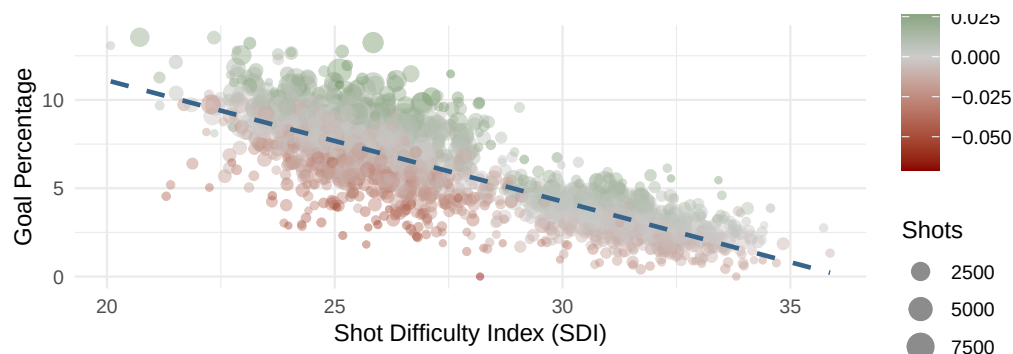


Figure 1: Figure 1. NHL Shot Difficulty Index vs Goal Percentage (players with ≥ 100 shots)

Figure 1 provides a preliminary check that harder average shot profiles trend toward lower conversion, while residual color-coding highlights potential over- and under-performers relative to xG .

Group Considerations

Our three-person team split work across data access, modeling, and communication deliverables so each league could be developed in parallel and then integrated in one document.

Task	Kanaan	Kaleb	Drianna
Data Collection (NBA/NHL)	0%	50%	50%
Feature Engineering + SDI	0%	50%	50%
Modeling (xFG/xG , diagnostics)	30%	40%	30%
Figures + Proposal Writing	33%	33%	33%